

Human-Computer Interaction

Qualitative Field Methods

Professor Bilge Mutlu

Today's Agenda

- When to go to the field: nascent theory & fresh looks
- Ethnography: settings, fieldwork, data
- Reading ethnography (Dourish): insight, not requirements
- Probes & diaries: studying life *in the wild*
- Hands-on: *field method for your question*, then a project check-in

Due today, 9:30 am

- **Design-led / WoZ Study Plan** (*individual*)
- **Research Questions + Literature Review** (*team milestone*)

Both on Canvas. Today's worksheet builds on the RQ you just submitted.

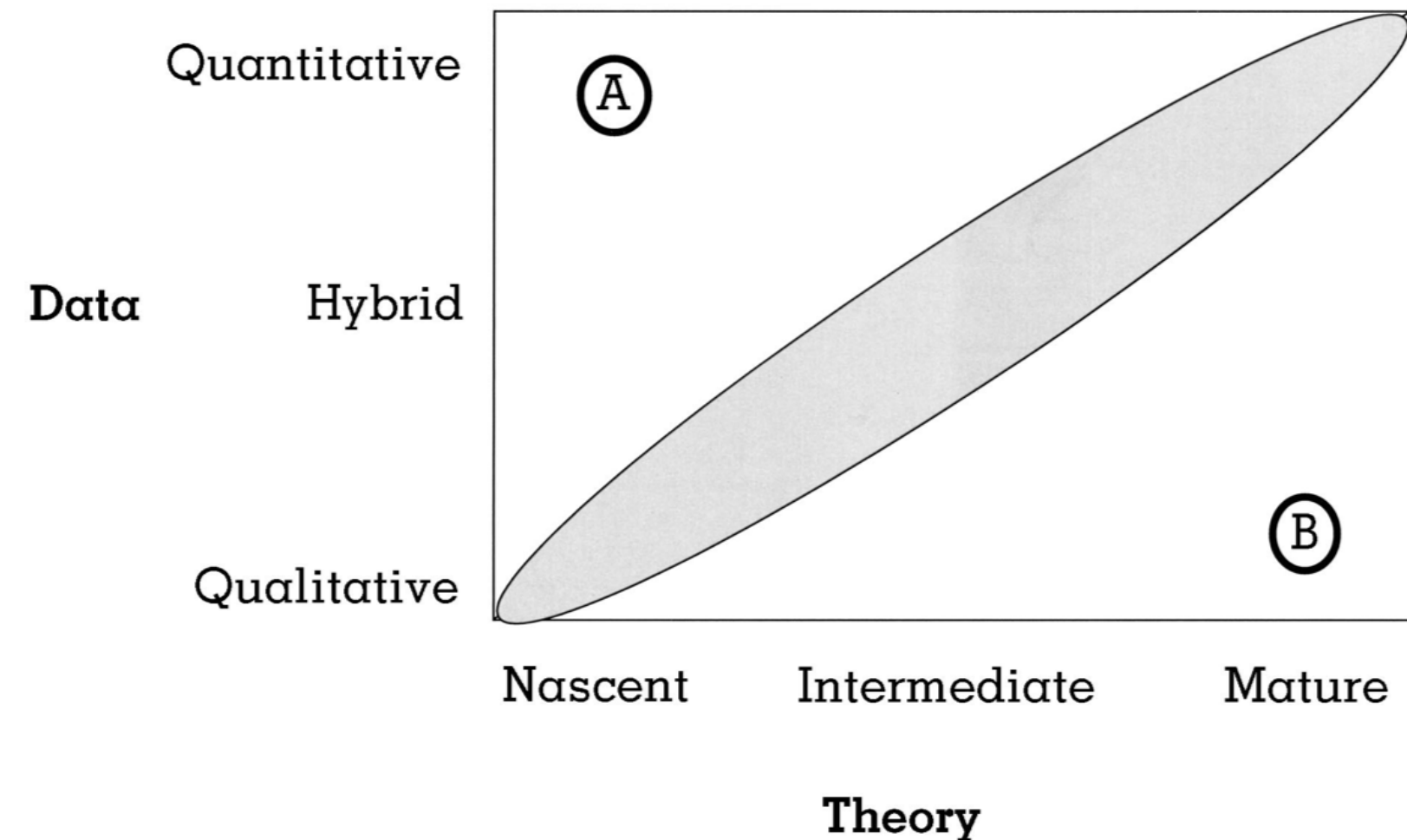
This week's readings

- **Required**
 - Dourish, *Reading and Interpreting Ethnography* (2014)
 - Gaver, Dunne & Pacenti, *Design: Cultural Probes* (1999)
- **Recommended:** LFH **Ch. 9 (Ethnography)** + **Ch. 6 (Diaries)**
- **Optional:** Consolvo & Walker, *Using ESM to Evaluate Ubicomp Applications* (2003)

When do we go to the field?

1. Theory is **nascent** in an area
2. You want to take a **fresh look** at a mature topic

FIGURE 1
Methodological Fit As a Mean Tendency



The qualitative family¹

Traditions: <i>the why</i>	Field methods: <i>the how</i>
Narrative (inquiry)	Observation & interviews
Phenomenology	Probes: cultural, technology
Grounded theory	Diaries & ESM
Case study	Day reconstruction
(Participatory) action research	
Ethnography ← today, part 1	↑ today, part 2

¹ Items 1-5: Creswell et al. (2007), *The Counseling Psychologist*, 35(2), 236-264.

Ethnography²

Definition: deep immersion and participation in a specific research context to develop an understanding not achievable with more limited approaches.

- Roots in anthropology & sociology, adopted by HCI
- Observation + interview data → new understanding and theory

²Lazar, Feng & Hochheiser (2017). *Research Methods in HCI*, Ch. 9 (Ethnography).

Where do we start?

1. Identify a domain where theory is *nascent* and new theory can have great impact
2. Develop a general *research question* and focus that can be updated
3. Find a *setting* to study the phenomena of interest
4. Conduct *fieldwork*, asking: **"What is going on here?"**

Setting & fieldwork

Setting: a site where the phenomena of interest can be observed.

Fieldwork: an organic process where data *collection* and *analysis* develop symbiotically, becoming more focused over time.

- Obtaining access, informed consent
- Identifying *stakeholders*, choosing a *role*
- Discovering groups & situations for comparison
- Writing detailed *fieldnotes*

How do we collect data?

1. **Fly-on-the-wall observation**: observe without influencing the context
 2. **Participant observation**: understand by taking part in the activities
 3. **Interviews**: understand perspectives; validate observations
- Each produces *fieldnotes* or *transcripts* (written, audio, video).

Reading ethnography³

Requirements are **one** use of ethnography, not the main one.

- Data is **generated** by taking part, not found lying around
- Two kinds of claims: **empirical** (what happened) and **conceptual** (a new way to understand it)
- Its design value is in **reframing** the questions of design

³ Dourish, P. (2014). Reading and Interpreting Ethnography. In *Ways of Knowing in HCI*, Springer, pp. 1-23.

Dourish: questions for an ethnography

Good questions

What are its **empirical** claims? Its **conceptual** claims?

How did the ethnographer **take part**?

How does it **add to** the body of work?

Less good questions

Is this a representative sample?

How can you tell if what people told you is right?

What should I build now that I know this?

Digital ethnography (netnography)⁴

- Extends ethnographic methods to online settings: forums, social media, virtual worlds
- Combines participant observation, interviews, analysis of digital artifacts
- Unique *ethical issues*: consent in online spaces, trace-data use
- Often *multi-sited*: tracing online↔offline practices

⁴Kozinets (2015). *Netnography: Redefined*. Sage. · Boellstorff et al. (2012). *Ethnography and Virtual Worlds*.

*But how do we study life in
the wild, asynchronously and
over time?*

Why probes & diaries?

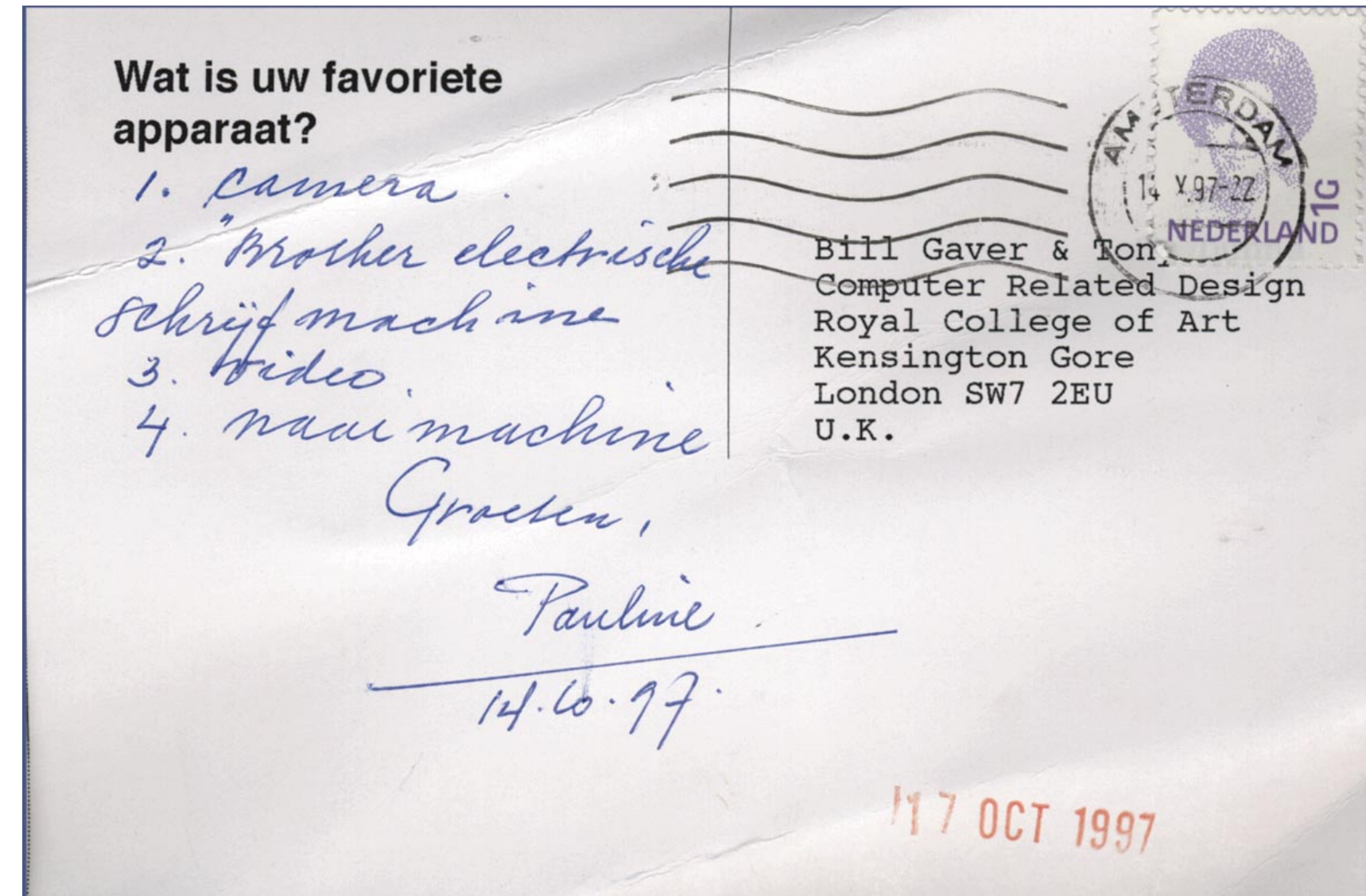
- **Challenge:** study technology use in *naturalistic, everyday contexts* where labs fall short
- **Goal:** capture rich, contextualized, temporal, and personal insight



Cultural probes⁵

Kits of evocative materials left with participants: postcards, maps, a disposable camera, a photo album, a media diary.

- **Inspiration, not information:** provoke design ideas, not measure variables
- Open-ended, playful, aesthetic; center participants' voices & cultural context



⁵ Gaver, Dunne & Pacenti (1999). Design: Cultural Probes. *interactions*, 6(1), 21-29.

Technology probes⁶

Simple technologies deployed in homes, serving three goals at once:

- *Social science*: understand needs in real contexts
- *Engineering*: field-test technology
- *Design*: inspire new ideas

Not hi-fi prototypes: deliberately simple, co-adaptive; collect logs of *actual use*.



⁶ Hutchinson et al. (2003). Technology Probes: Inspiring Design for and with Families. *CHI 2003*.

Diaries & experience sampling

- **Diary:** self-report log of events, feelings, experiences over time
- **Experience Sampling (ESM):** record at *random or prompted* times (beepers → smartphones)

Capture experiences *as they happen* → reduce recall bias; cover multiple contexts.

Day reconstruction method (DRM)⁷

An alternative to ESM: reconstruct **yesterday**, episode by episode.

1. Write a short diary of the previous day as a sequence of episodes
2. For each episode: when, what, where, with whom
3. Rate how you felt (12 feelings, 0–6)

Less burden than ESM, no interruptions, and the **whole day** instead of sampled moments.

⁷Kahneman, D., Krueger, A.B., Schkade, D.A., Schwarz, N. & Stone, A.A. (2004). A Survey Method for Characterizing Daily Life Experience: The Day Reconstruction Method. *Science*, 306(5702), 1776–1780.

What a DRM day looks like

	Mean affect rating					Mean hours/day	Proportion of sample reporting
	Positive	Negative	Competent	Impatient	Tired		
Activities							
Intimate relations	5.10	0.36	4.57	0.74	3.09	0.2	0.11
Socializing	4.59	0.57	4.32	1.20	2.33	2.3	0.65
Relaxing	4.42	0.51	4.05	0.84	3.44	2.2	0.77
Pray/worship/meditate	4.35	0.59	4.45	1.04	2.95	0.4	0.23
Eating	4.34	0.59	4.12	0.95	2.55	2.2	0.94
Exercising	4.31	0.50	4.26	1.58	2.42	0.2	0.16
Watching TV	4.19	0.58	3.95	1.02	3.54	2.2	0.75
Shopping	3.95	0.74	4.26	2.08	2.66	0.4	0.30
Preparing food	3.93	0.69	4.20	1.54	3.11	1.1	0.62
On the phone	3.92	0.85	4.35	1.92	2.92	2.5	0.61
Napping	3.87	0.60	3.26	0.91	4.30	0.9	0.43
Taking care of my children	3.86	0.91	4.19	1.95	3.56	1.1	0.36
Computer/e-mail/Internet	3.81	0.80	4.57	1.93	2.62	1.9	0.47
Housework	3.73	0.77	4.23	2.11	3.40	1.1	0.49
Working	3.62	0.97	4.45	2.70	2.42	6.9	1.00
Commuting	3.45	0.89	4.09	2.60	2.75	1.6	0.87

Choosing the method

Method	Purpose	When to use
Cultural probes	Inspire design ideas	Early design exploration
Technology probes	Explore + co-design	Early/mid design, field testing
Diaries / ESM	Document experiences	Practices & behaviors over time
Day reconstruction	Reconstruct a whole day	Daily experience, low burden

Decision guide: inspiration → cultural probe · field test + ideas → technology probe · in-the-moment data → diary / ESM · whole day, less burden → DRM

Hands-on activity (30 min)

"Field method for your question"

With your team, take your project and design a qualitative field pass:

- (12 min) Choose. **Ethnographic observation**, a **probe** (cultural or technology), or a **diary / ESM / DRM** study for one open part of your RQ, and why it fits
- (18 min) Draft. The actual instrument: observation guide, probe kit, or diary prompts, plus what you'd capture

Grab the worksheet

- One copy per team: File → Make a copy
- Nothing to submit today
- Each of you turns it into your **Qualitative Field-Method Plan**

da.gd/EVHaO



Project check-in (10 min)

Each team, in one breath:

- Your **method family**: field, design-led, or experimental
- **One open design question** you haven't settled

Assignment: Qualitative Field-Method Plan

Due: Fri **Oct 9, 9:30 am** · $\approx$ 1–2 pp, individual, on Canvas

1. **Method choice + rationale:** why it fits your RQ and access
2. **Instrument:** observation guide, probe kit, or diary prompts
3. **Sampling & access:** who, how recruited, over what timescale
4. **Analysis intent:** what you'll do with the data
5. **Reflection:** strengths, weaknesses, what you'd change

Rubric ($\checkmark$ +/ $\checkmark$ / $\checkmark$ -): completeness · fit · feasibility

What's Next?

- **Monday: Generative AI & Tools for Thought**
 - **Read:** Lee et al. (CHI 2025) · **If you can:** Bush (1945)
 - Post your provocation slide
- **Due Fri Oct 9, 9:30 am:** Field-Method Plan (*individual*)
- **Method design** (draft) → Method-Design Crit, Oct 16